

Venkatesh R

Full Stack Developer | React.js · Spring Boot · Python · Machine Learning

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LinkedIn

GitHub

Portfolio

PROFESSIONAL SUMMARY

Results-driven **Full Stack Developer** with **2.2+ years** of professional experience building scalable, AI-powered web applications using **React.js**, **Spring Boot**, and **Python**. Specialized in developing **algorithmic trading systems** powered by **XGBoost ML models**, real-time data processing, and automated trade execution. Proficient in designing **RESTful APIs**, **JWT-based authentication**, and responsive **UI/UX** with modern front-end technologies. Hands-on with **DevOps**, **AWS cloud deployments**, and **ML model integration** for production-grade financial applications. Strong collaborator in Agile environments focused on delivering optimized, maintainable, and high-performance solutions.

TECHNICAL SKILLS

Frontend: React.js, JavaScript (ES6+), TypeScript, Redux Toolkit, React Hooks, HTML5, CSS3, Tailwind CSS, Bootstrap, Material-UI, Chart.js, Responsive Design

Backend: Spring Boot, Java 17/21, Python, Flask, FastAPI, RESTful APIs, Microservices, Spring Data JPA, Hibernate, Spring Security, JWT Authentication, Node.js

AI / ML & Data Science: XGBoost, Scikit-learn, Pandas, NumPy, TensorFlow, Feature Engineering, Time-Series Analysis, Model Training & Optimization, Backtesting

Databases: MySQL, MongoDB, PostgreSQL, Redis, InfluxDB, Database Design, Query Optimization, Time-Series Storage

DevOps & Cloud: AWS (EC2, S3, RDS, Lambda), Docker, Jenkins, GitHub Actions, CI/CD, Render, Vercel, Nginx, WebSocket, Apache Kafka

Tools & Trading APIs: Git, GitHub, Postman, Maven, pip, IntelliJ IDEA, VS Code, PyCharm, Angel One SmartAPI, Zerodha Kite API, Jira, Agile / Scrum

PROFESSIONAL EXPERIENCE

Full Stack Developer

MAK Infotech Pvt. Ltd.

Dec 2023 – Present

Dharmapuri, Tamil Nadu

- Joined full-time after completing professional certifications post-graduation; leading end-to-end development of an **AI-Powered Algorithmic Trading Platform**.
- Spearheading an **XGBoost-based trading engine** integrated with **Spring Boot microservices** and **Python ML pipelines** for automated signal generation.
- Built and deployed **ML models** for stock prediction using **XGBoost**, achieving **78% accuracy** on intraday price movement classification.

- Developed responsive, real-time trading dashboards in **React.js** with **WebSocket** integration for live market data and P&L visualization.
- Engineered secure **Spring Boot REST APIs** integrated with **FastAPI microservices** for sub-50 ms ML inference.
- Improved system throughput by **50%** through Redis caching, database indexing, and optimized feature-engineering pipelines.
- Implemented **JWT authentication**, **Spring Security**, and role-based access control for multi-tenant trading environments.
- Designed automated **backtesting frameworks** processing **5+ years of historical data** to validate trading strategies (Sharpe ratio ≥ 1.8).
- Managed CI/CD pipelines and cloud deployments using **GitHub Actions**, **Docker**, and **AWS EC2**, maintaining **99.9% uptime**.

Software Developer Intern

MAK Infotech Pvt. Ltd.

Jun 2022 – May 2023

Dharmapuri, Tamil Nadu

- Completed a **1-year internship** during M.Sc. final year, contributing to a **Mobile E-Commerce Platform** serving 10,000+ SKUs.
- Developed responsive UI components using **React.js** and **Tailwind CSS** for a mobile-first e-commerce experience.
- Built **Spring Boot REST APIs** for product catalog, user authentication, and order management modules.
- Implemented **JWT-based authentication** and role-based access control for admin, vendor, and customer portals.
- Integrated **Redis caching** for product listings and sessions, reducing API latency by **45%**.
- Gained hands-on expertise in **Git**, **Agile/Scrum**, code reviews, and **clean code architecture**.

KEY PROJECTS

AI-Powered Algorithmic Trading Platform

Dec 2023 – Present

Python · XGBoost · Spring Boot · React.js · FastAPI · WebSocket · Redis · MongoDB · Angel One SmartAPI

- Architected a comprehensive **algorithmic trading system** with an **XGBoost prediction engine** for automated trade execution across top 500 NSE stocks.
- Built a **Python ML pipeline** featuring technical indicators (RSI, MACD, Bollinger Bands, VWAP) and market-microstructure features.
- Implemented intelligent **QML signal detection**, **Break of Structure (BOS)**, and **CHOCH** logic fused with ML predictions.
- Developed a **FastAPI microservice** for real-time inference with sub-50 ms latency, bridged to Spring Boot backend.
- Created a **live trading engine** with dynamic capital allocation, risk management, TP/SL logic, and automated square-off.
- Integrated with **Angel One SmartAPI** and **Zerodha Kite API** for multi-broker execution.
- Deployed on **AWS EC2** with Docker containers and CI/CD automation.

Mobile E-Commerce Platform

Jun 2022 – May 2023

React.js · Tailwind CSS · Spring Boot · Redis · MySQL · JWT · Razorpay · AWS

- Developed a full-featured **mobile-first e-commerce platform** for smartphones and electronics with complete order lifecycle management.
- Designed **product catalog** with advanced search, filtering, and category management for 10,000+ SKUs.
- Built **shopping cart, wishlist, and checkout modules** with **Razorpay** payment integration.
- Implemented **inventory management** with real-time stock updates and low-stock alerts.
- Developed an **admin dashboard** with sales analytics using **Chart.js** visualizations.
- Enhanced performance by **40%** via lazy loading, code splitting, and optimized API calls.

EDUCATION

M.Sc. in Computer Science

Aug 2021 – May 2023

Government Arts and Science College, Pennagaram

CGPA: 7.43 / 10

- Pennagaram, Tamil Nadu | *Advanced Algorithms, Cloud Computing, Machine Learning, Distributed Systems*

B.Sc. in Computer Science

Jul 2018 – Jun 2021

B.R.P.T. Dr. M.G.R. Govt. Arts and Science College, Palacode

CGPA: 7.34 / 10

- Palacode, Tamil Nadu | *Data Structures, DBMS, Web Technologies, Operating Systems, OOP*

ACADEMIC PROJECT

Human Face Emotion-Based Music Player

M.Sc. Final Year

Python · TensorFlow · OpenCV · CNN · HTML5 · CSS3 · JavaScript

- Developed an AI-powered music player that detects facial emotions in real time using **TensorFlow** and **OpenCV**.
- Trained a **Convolutional Neural Network (CNN)** for accurate emotion recognition on a custom dataset.
- Designed a responsive interface using **HTML5, CSS3, and JavaScript** integrated with Python backend.
- Source Code: [GitHub Repository](#)

CERTIFICATIONS & PROFESSIONAL TRAINING

Completed post-graduation to strengthen full-stack, cloud, and ML expertise before joining full-time role.

- **Java Full Stack Development — Spring Boot, React.js, REST APIs, SQL & AWS Deployment** *Jun – Jul 2023*
Comprehensive training covering Spring Boot microservices, React.js SPA development, REST API design, SQL optimization, and AWS EC2/S3 deployment workflows.
- **Python Advanced Programming — Data Handling, Automation & Backend Scripting** *Jul – Aug 2023*
In-depth Python course covering Pandas, NumPy, scripting automation, Flask/FastAPI backends, and data pipeline development.
- **Machine Learning with Python — XGBoost, Scikit-learn, Feature Engineering & Model Deployment** *Sep – Oct 2023*
Hands-on ML training covering supervised/unsupervised algorithms, XGBoost tuning, feature engineering, model evaluation, and production deployment using FastAPI.
- **AWS Cloud Practitioner Essentials — Cloud Fundamentals, EC2, S3, RDS & Lambda** *Nov 2023*
AWS foundational course covering cloud architecture, compute, storage, database services, security best practices, and cost optimization.